Multiscale Latent Geometry for Quantitative Functional Lung Imaging

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Abstract

Quantification of hyperpolarized pulmonary ventilation MRI relies on scalar summaries such as ventilation defect percentage, which omit consideration of the spatial organization of functional abnormalities. We introduce a homeomorphic multiscale latent geometry for spatially informed analysis of pulmonary ventilation, constructed using trained three-dimensional normalizing flow (i.e., Glow) networks. Each ventilation volume is mapped through a continuous bijection (with a continuous inverse) to a Gaussian latent distribution. The trained network decomposes each image into multiscale resolution levels, facilitating complementary characterization of local and global ventilation patterns without loss of information. The resulting structured latent representation supports comparative quantification through pairwise distances and spherical interpolation trajectories within the Gaussian typical set. Moreover, the homeomorphic construction preserves topological neighborhoods between the image and latent spaces, such that local perturbations and continuous trajectories in image space correspond to continuous changes in the latent space. Relationships can therefore be decoded as complete ventilation volumes for direct visual and quantitative interrogation. We evaluated the framework in an exploratory cohort of 45 participants comprising cystic fibrosis, chronic obstructive pulmonary disease, interstitial lung disease, and young and older healthy groups. The learned geometry exhibited scale-dependent organization associated with clinical phenotypes. Fine and coarse latent variables showed complementary group relationships consistent with localized abnormalities and global ventilation organization, respectively. These findings establish the feasibility of latent modeling for functional lung MRI and provide a continuous, spatially informed alternative to one-dimensional, defect-based summaries. The framework supports unsupervised phenotyping, multiscale quantification, and generative interrogation of pulmonary ventilation.

Keywords: functional lung imaging; hyperpolarized xenon-129 MRI; normalizing flows; homeomorphism; latent geometry; unsupervised phenotyping

1 Introduction

1.1 Quantitative functional imaging of pulmonary ventilation

The lung presents a distinctive challenge for magnetic resonance imaging (MRI). Its low proton density, rapid transverse signal decay, and numerous air–tissue interfaces substantially limit conventional proton-based imaging of regional pulmonary function. Hyperpolarized noble gases provide a means of circumventing these limitations by directly visualizing the distribution of an inhaled contrast agent within the airspaces of the lung. Early studies using hyperpolarized helium-3 demonstrated the feasibility of depicting normal and abnormal ventilation and established the sensitivity of regional signal abnormalities to obstructive lung disease (Bachert et al., 1996; Kauczor et al., 1997, 1996). Subsequent development of hyperpolarized xenon-129 (^{129}Xe) MRI extended this capability while offering practical advantages in availability and the additional potential to characterize gas exchange (Dregely et al., 2011; Mugler et al., 2010). Ventilation imaging with hyperpolarized gas has since been applied across a broad range of pulmonary conditions, including asthma, cystic fibrosis (CF), chronic obstructive pulmonary disease (COPD), and interstitial lung disease (ILD) (Altes et al., 2001; Kirby et al., 2012b; Mammarappallil et al., 2019; Myc et al., 2020; Santyr et al., 2019). Recent reviews summarize the expanding set of quantitative ventilation, gas-exchange, and microstructural measurements (Stewart et al., 2022), including their emerging roles across cardiopulmonary disease (Schmidt et al., 2025).

The principal image feature used in many of these applications is the ventilation defect. A ventilation defect is operationally defined as a lung region exhibiting absent or abnormally low hyperpolarized gas signal, consistent with nonventilated or underventilated airspaces. Relatedly, ventilation defect percentage (VDP) expresses the corresponding defect volume as a percentage of total lung volume (Niedbalski et al., 2021). Although such defects can be visually conspicuous, their conversion into reproducible quantitative measurements requires decisions concerning lung delineation, intensity normalization, and assignment of image voxels to ventilation classes. A variety of algorithms have therefore been proposed, including binary thresholding, linear binning, hierarchical and adaptive clustering, fuzzy spatial clustering, and probabilistic mixture modeling (He et al., 2020, 2016; Hughes et al., 2018; Kirby et al., 2012a; Tustison et al., 2011; Zha et al., 2016). More recently, convolutional neural networks have enabled direct image-domain segmentation while incorporating spatial context unavailable to intensity-only approaches (Tustison et al., 2021, 2019).

VDP is intuitive, clinically interpretable, and has demonstrated repeatability and potential utility as an imaging biomarker in single- and multisite studies (Couch et al., 2019; Svenningsen et al., 2020). Nevertheless, the transformation from a three-dimensional ventilation image to a single percentage is necessarily lossy. Subjects with different numbers, sizes, locations, and spatial distributions of defects can have the same VDP. The measurement consequently preserves overall defect burden while discarding much of the spatial information that distinguishes diffuse from focal abnormality, peripheral from central involvement, and regionally organized from spatially dispersed dysfunction. These distinctions are potentially important because pulmonary diseases may produce overlapping global burdens while differing in their regional expression and underlying pathophysiology.

Evidence that ventilation images contain clinically relevant information beyond global defect burden predates recent deep learning approaches. In an early analysis of hyperpolarized 3He ventilation MRI, approximately 1,600 image-derived features were evaluated in 55 participants with and without asthma (Tustison et al., 2010). The highest-ranked image features individually carried substantially more information about diagnostic status than standard spirometric measurements, while the imaging and spirometric features provided largely complementary information. This study demonstrated the phenotypic value of spatially resolved ventilation patterns, but it relied on a large set of predefined, handcrafted descriptors. More recent studies have extended this observation using spatial distribution indices, texture analysis, and learned image representations. A three-dimensional defect distribution index distinguished the spatial clustering of ventilation defects across obstructive and restrictive disease groups, even when interpreted alongside VDP (Bdaiwi et al., 2025). Texture features derived from ^{129}Xe ventilation MRI were also associated with longitudinal quality-of-life improvement in long COVID (Kooner et al., 2024). Most recently, a supervised convolutional network classified pulmonary diseases from ventilation images with performance exceeding that of human observers, supporting the presence of disease-associated spatial patterns not represented by VDP alone (Matheson et al., 2025).

This limitation is related to, but more fundamental than, the distinction between histogram- and image-based segmentation. We previously showed that histogram-based quantification discards contextual spatial information and can exhibit reduced measurement precision under common MRI perturbations relative to direct image-domain analysis (Tustison et al., 2021). An image-based segmentation model mitigates part of this loss by using spatial information to assign labels. However, the final conversion of that segmentation to VDP again collapses the result to a scalar. Thus, even an accurate segmentation does not by itself provide a representation of how ventilation abnormalities are spatially organized across subjects. A more general quantitative framework would retain the complete image, support comparisons at multiple spatial scales, and provide a principled geometry for measuring population variation without requiring predefined ventilation classes.

1.2 Invertible generative modeling as a quantitative framework

Deep generative models provide a possible route from scalar quantification toward population-level representation of complete images. In contrast to discriminative models optimized for a specific label or endpoint, generative models attempt to characterize the distribution from which the observations arise. This distinction is useful for functional lung imaging, where the desired representation may need to support several downstream tasks, including phenotyping, similarity analysis, interpolation, outlier detection, and eventually conditional inference, without committing the training procedure to a single clinical categorization.

Normalizing flows are particularly attractive for this purpose because they define a bijective transformation between the image domain and a tractable latent probability distribution (Dinh et al., 2017; Kingma and Dhariwal, 2018; Kobyzev et al., 2021; Papamakarios et al., 2021). Unlike generative adversarial networks, which learn an implicit distribution, or conventional variational autoencoders, which generally trade exact reconstruction for a lower-dimensional stochastic code, normalizing flows retain an explicit inverse and permits exact likelihood evaluation under the specified model. Each observed image can therefore be mapped to a unique latent representation and reconstructed from that representation, subject to numerical precision. This correspondence establishes a coordinate system in which the variability of the observed population can be examined quantitatively (Tustison et al., 2026).

Flow-based modeling of pulmonary ventilation nevertheless presents several technical challenges. Hyperpolarized gas ventilation MR images contain extensive background regions, with informative signal largely confined to the lungs and distributed heterogeneously within the pulmonary volume. To improve robustness and reduce sensitivity to nearly uniform background intensities, we introduce small random intensity and shape perturbations during training as a form of data augmentation (Tustison et al., 2025, 2024; Tustison et al., 2021). The perturbation magnitude provide sufficient regularization without obscuring clinically meaningful regional ventilation patterns. In addition, full three-dimensional modeling imposes substantial memory and computational requirements, making the balance among spatial resolution, multiscale depth, and network capacity especially consequential. We refer to this augmentation as "dequantization noise" for consistency with flow-modeling terminology (i.e., (Ho et al., 2019)), although here it is used primarily as a data smoothing operation rather than to correct an assumed discrete data-generating process.

The multiscale construction introduced by Glow offers a natural organization for medical and biological image data (Kingma and Dhariwal, 2018). Successive squeezing, invertible transformation, and splitting operations factor the image representation across resolution levels. Rather than interpreting the latent variables as a single undifferentiated vector, the resulting hierarchy can be interrogated by resolution level to determine how group relationships change across spatial scales. Pretrained vision and vision-language foundation models provide an alternative means of encoding medical images into compact representations for similarity analysis, clustering, content-based retrieval, and downstream prediction (Codella et al., 2024; Denner et al., 2025; Zhang et al., 2025). Such embeddings can capture transferable semantic features, but they are generally optimized for invariance or discrimination rather than information-preserving. By contrast, the flow-based mapping remains invertible, allowing individual latent levels to be manipulated and decoded to assess their contributions in the image domain. This combination of multiscale organization and exact invertibility distinguishes the proposed analysis from both post hoc dimensionality reduction and foundation-model embeddings, for which correspondence with the complete image is generally noninvertible.

An additional consideration concerns the geometry of the Gaussian latent distribution. In high dimensions, probability mass is concentrated within a typical set located away from the origin. Consequently, the Gaussian mean is a mathematically convenient reference but is not representative of a typical encoded subject. Linear interpolation through the origin can similarly traverse low-probability regions and produce trajectories that are inconsistent with the radial organization of the latent distribution. Normalizing latent representations to a common radius and using spherical linear interpolation provides an alternative that follows the hyperspherical typical set. Pairwise angular or arc-length distances along this set can then be evaluated separately at each resolution level or combined across the complete latent representation. Importantly, these distances describe geometry induced by the learned model and their biological meaning must be established empirically rather than assumed from the Gaussian prior alone (Arvanitidis et al., 2018; White, 2016).

1.3 Motivation

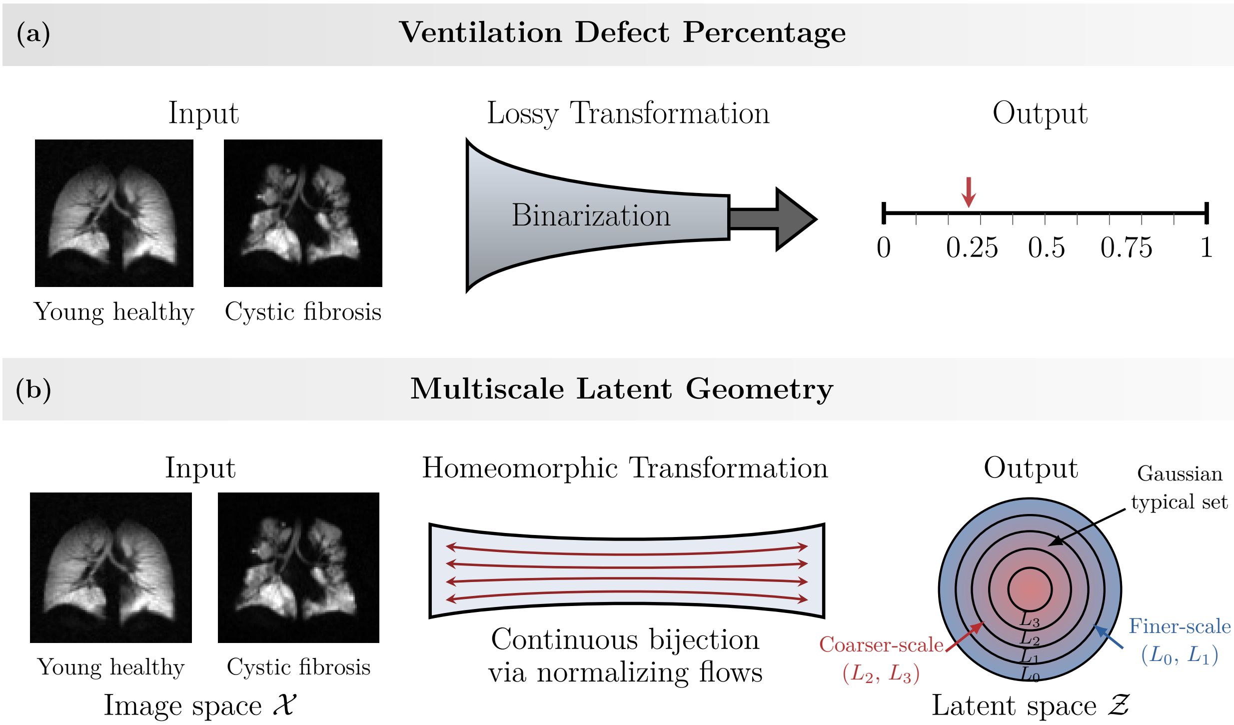

Figure 1: Scalar reduction versus invertible multiscale representation of pulmonary ventilation. Representative ^{129}Xe ventilation MR images from a young healthy participant and a participant with cystic fibrosis are shown to illustrate the two analysis pathways. (a) Conventional ventilation defect percentage (VDP) analysis requires binarization of the ventilation image into defect and nondefect classes followed by calculation of the fraction of lung assigned to the defect class. This lossy transformation preserves global defect burden but discards the number, size, location, and spatial arrangement of the defects. (b) The proposed normalizing flows framework learns a continuous bijection between image space $\mathcal{X}$ and latent space $\mathcal{Z}$. The bidirectional streamlines denote the forward and inverse transformations, through which each image is encoded and reconstructed. The multiscale architecture factors the latent representation across levels L_0–L_3, enabling finer- and coarser-scale contributions to population geometry to be examined separately or jointly. The concentric latent-space depiction schematically represents the Gaussian typical set and the multiscale organization.

Figure 1 summarizes the central methodological contrast motivating this study. Conventional VDP quantification maps a complete ventilation image, through binarization, to a single scalar value denoting the global defect fraction. Our framework instead maps the image bijectively to a multiscale latent representation, retaining the information required for reconstruction while providing coordinates in which population variation can be analyzed. Motivated by this contrast, we investigate normalizing flows as a general quantitative framework for three-dimensional hyperpolarized ^{129}Xe ventilation MRI. Our objective is not to replace a specific VDP segmentation algorithm or to optimize diagnostic classification. Instead, we ask whether an invertible, multiscale density model can preserve pulmonary ventilation images in a continuous latent coordinate system whose geometry contains clinically relevant structure despite being learned without diagnostic supervision. This formulation treats disease labels as external variables for evaluating the learned representation rather than targets used to construct it.

We make four principal contributions. First, we adapt a multiscale normalizing flow architecture to the low-resolution, spatially heterogeneous characteristics of hyperpolarized gas ventilation MRI through shape- and intensity-based data augmentation. Second, we define subject-level representations from the learned flow latents and perform comparisons within the Gaussian typical set, enabling pairwise distance analysis and spherical interpolation. Third, we exploit the multiscale factorization of the flow to characterize complementary phenotypic information at individual latent resolution levels and across their combination, while retaining the ability to decode level-specific latent manipulations into the image domain. Fourth, we evaluate the learned geometry in an exploratory cohort comprising young and older healthy volunteers, together with participants with CF, COPD, or ILD. Because diagnostic group labels are not used during training, their correspondence with the latent organization provides an independent, post hoc assessment of whether the unsupervised model captures clinically meaningful variation in pulmonary ventilation. Notably all functionality is publicly available as mature open-source software via the ANTsX ecosystem.[1]

[1] https://github.com/ANTsX

2 Materials and methods

2.1 Study cohort

This retrospective exploratory study comprised 45 participants with hyperpolarized xenon-129 (^{129}Xe) pulmonary ventilation MRI. The cohort included young healthy volunteers, older healthy volunteers, and participants with cystic fibrosis (CF), chronic obstructive pulmonary disease (COPD), or interstitial lung disease (ILD). Diagnostic labels were used only for evaluation of the learned representation and were not provided to the normalizing-flow model during training.

TODO: Provide the number of participants in each group; demographic characteristics; inclusion and exclusion criteria; recruitment source; institutional review board approval; consent or waiver-of-consent language; and the criteria used to assign clinical diagnoses.

2.2 Hyperpolarized xenon-129 MRI

Only hyperpolarized ^{129}Xe ventilation images were used. No paired proton structural image, dissolved-phase ^{129}Xe image, clinical measurement, or diagnostic variable was supplied to the model. Each subject was represented by a single three-dimensional ventilation volume in NIfTI format.

TODO: Provide the scanner manufacturer and model, field strength, radiofrequency coil, pulse sequence, acquisition dimensionality, field of view, acquired voxel dimensions, repetition and echo times, flip angle, breath-hold protocol, xenon polarization and dose, reconstruction procedure, and any acquisition differences across participants. If the images originated from more than one protocol or site, specify these factors and whether they were balanced across groups.

2.3 Image preprocessing

Each ventilation image was spatially resampled to a common matrix of $48 \times 32 \times 48$ voxels. Resampling provided a fixed-dimensional input compatible with the three-dimensional invertible architecture while retaining the complete ventilation volume. The resulting images were subjected to probabilistic dequantization before likelihood-based training. Specifically, continuous noise with a configured scale of $\alpha = 0.15$ was added to the discretized intensities. Dequantization replaces point masses associated with repeated digital intensity values—particularly the extensive black background characteristic of hyperpolarized gas images—with local continuous distributions, thereby making the observations compatible with continuous-density estimation (Ho et al., 2019).

The same deterministic spatial and intensity preprocessing operations were applied to all subjects before assignment to the training or evaluation subsets. Diagnostic labels did not influence preprocessing.

TODO: Specify (1) the spatial coordinate system and interpolation method used for resampling; (2) whether the images were registered to a reference image or template; (3) the procedure for cropping or padding the lung field; (4) intensity clipping and normalization; (5) whether an explicit lung mask was used; (6) the distribution of the dequantization noise and whether it was applied during every training iteration or once before training; and (7) the treatment of dequantization during validation and inference.

2.4 Three-dimensional multiscale normalizing flow

2.4.1 Invertible transformation

Let $\mathbf{x} \in \mathbb{R}^D$ denote a vectorized, dequantized ventilation volume and let f_θ denote an invertible transformation parameterized by θ. The model maps the image to a latent variable

$$\mathbf{z} = f_\theta(\mathbf{x}),$$

with inverse reconstruction

$$\mathbf{x} = f_\theta^{-1}(\mathbf{z}).$$

A standard multivariate Gaussian distribution was used as the base density, $p_Z(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$. Through the change-of-variables formula, the image log likelihood was

$$\log p_X(\mathbf{x}) = \log p_Z(f_\theta(\mathbf{x})) + \log \left| \det \frac{\partial f_\theta(\mathbf{x})}{\partial \mathbf{x}} \right|.$$

The bijective construction permits direct likelihood evaluation and reconstruction of each image from its latent representation (Dinh et al., 2017; Kobyzev et al., 2021; Papamakarios et al., 2021).

2.4.2 Flow blocks and multiscale factorization

The model extended the Glow architecture to three spatial dimensions (Kingma and Dhariwal, 2018). Each flow step comprised activation normalization, an invertible $1 \times 1 \times 1$ convolution, and an affine coupling transformation. Within the coupling layer, the input channels were partitioned into two components. One component was passed unchanged, whereas a three-dimensional convolutional subnetwork predicted the scale and translation applied to the second component. This triangular construction provided an expressive nonlinear transformation while retaining an efficiently computable Jacobian determinant and analytic inverse.

The architecture used four resolution levels, denoted L_0 through L_3. At each level, a three-dimensional squeeze operation exchanged spatial resolution for channels, followed by a sequence of invertible flow steps. A portion of the channels was factored from the transformation at successive levels, producing the multiscale latent representation

$$\mathbf{z} = (\mathbf{z}_0, \mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3)\,.$$

This factorization allowed the complete image likelihood and reconstruction to be retained while enabling separate analysis of the latent variables associated with individual resolution levels. Because the current study used a single ventilation image per participant, no inter-view alignment loss or multimodal conditioning term was included.

TODO: Supply the number of flow steps per level, coupling-network hidden channels, scale-clamping rule, activation functions, parameter initialization, base-distribution parameterization, and total number of trainable parameters. Confirm whether all four levels used split operations and specify the exact dimensionality of each $\mathbf{z}_l$.

2.5 Model optimization and selection

Parameters were optimized by minimizing the negative log likelihood, reported in bits per dimension:

$$\mathrm{bpd}(\mathbf{x}) = -\frac{\log p_X(\mathbf{x})}{D \log 2}.$$

Training was entirely unsupervised with respect to the five clinical groups. Accordingly, neither the diagnostic labels nor any loss designed to increase between-group separation contributed to optimization. The selected checkpoint was determined from likelihood performance on data excluded from gradient-based training.

TODO: Specify the training, validation, and test allocation; whether splitting was stratified by group; random seed; optimizer; learning rate and schedule; warm-up duration; batch size and gradient accumulation; number of iterations or epochs; gradient clipping; exponential moving average; checkpoint-selection criterion; numerical precision; GPU model and number of GPUs; software versions; and approximate training time. Given the small cohort, also clarify whether a single split, repeated splits, or cross-validation was used.

2.6 Multiscale latent geometry

2.6.1 Typical-set representation

For a high-dimensional standard Gaussian distribution, typical samples concentrate within a relatively thin annular region rather than near the origin. The origin is therefore the mode of the density but is not representative of a typical encoded image. To avoid interpreting distance from $\mathbf{z} = \mathbf{0}$ as a population-normality score, the present analysis focused on relationships between subjects within the Gaussian typical set.

For subject i and level l, the flattened latent variable $\mathbf{z}_{il}$ was projected onto a hypersphere of common radius r_l:

$$\widetilde{\mathbf{z}}_{il} = r_l \frac{\mathbf{z}_{il}}{\|\mathbf{z}_{il}\|_2}.$$

TODO: State how r_l was selected—for example, the theoretical Gaussian typical radius $\sqrt{d_l}$, the empirical mean radius, or another estimate—and whether centering or covariance normalization preceded radial projection.

2.6.2 Pairwise distances and spherical interpolation

For subjects i and j, the angular separation at level l was

$$\theta_{ijl} = \arccos\left(\frac{\widetilde{\mathbf{z}}_{il}^{\mathsf{T}}\widetilde{\mathbf{z}}_{jl}}{\|\widetilde{\mathbf{z}}_{il}\|_2\|\widetilde{\mathbf{z}}_{jl}\|_2}\right).$$

The corresponding hyperspherical arc length was $d_{ijl} = r_l\theta_{ijl}$. Spherical linear interpolation (Slerp) between the two latent representations was defined for $t \in [0, 1]$ as

$$\mathrm{Slerp}(\widetilde{\mathbf{z}}_{il}, \widetilde{\mathbf{z}}_{jl}; t) = \frac{\sin((1-t)\theta_{ijl})}{\sin(\theta_{ijl})}\widetilde{\mathbf{z}}_{il} + \frac{\sin(t\theta_{ijl})}{\sin(\theta_{ijl})}\widetilde{\mathbf{z}}_{jl}.$$

Intermediate latent variables could be decoded through f_θ^{-1} to visualize image-domain changes along the trajectory. Distances were calculated independently for L_0–L_3. A total multiscale distance summarized the level-specific distances for each subject pair.

TODO: Verify that the implemented distance is the hyperspherical arc length described above rather than the chord distance, an unscaled angular distance, or the numerical length of sampled Slerp points. Provide the exact formula used to combine $d_{ij0}, \ldots, d_{ij3}$ into the total distance, including any normalization for the unequal latent dimensionalities.

2.7 Evaluation of clinical organization

The clinical groups were used after training to evaluate whether the unsupervised latent representation contained clinically relevant organization. For each resolution level and for the combined representation, the complete 45×45 subject-distance matrix was constructed. Evaluation concerned two related properties: whether group membership was associated with systematic location differences in the latent geometry and whether the groups differed in within-group dispersion.

An omnibus permutational multivariate analysis of variance (PERMANOVA) was planned for each of the five distance matrices. Statistical significance was estimated by permuting the clinical labels at the subject level while leaving the distance matrix fixed. Because each subject contributes to multiple pairwise distances, the subject rather than the individual distance was the exchangeable unit. Permutational analysis of multivariate dispersion (PERMDISP) was used as a complementary analysis to determine whether a significant PERMANOVA result could be attributable to unequal group dispersion rather than differences in group location.

Following the omnibus analysis, pairwise comparisons were performed for the ten unique pairs formed by the five clinical groups. Applying these comparisons to the four resolution levels and the total distance yielded 50 planned pairwise tests. Family-wise error was controlled using Bonferroni correction, giving $\alpha_{\text{adj}} = 0.05/50 = 1.0 \times 10^{-3}$. For each comparison, an effect-size estimate and the groupwise within- and between-subject distance summaries were reported together with the permutation-derived p value.

TODO: Implement or confirm the subject-level PERMANOVA and PERMDISP analyses; specify the software, pseudo-F statistic, number of permutations, restricted-permutation scheme if applicable, effect-size definition, and confidence-interval procedure. The earlier Welch tests applied to individual pairwise distances should not be reported because pairwise distances sharing subjects are statistically dependent.

2.8 Software and reproducibility

The three-dimensional flow model, preprocessing, latent encoding, image reconstruction, and trajectory analysis were implemented within the ANTsX/ANTsTorch software ecosystem. The complete framework was designed to use publicly available, scriptable components to facilitate reproduction and extension of the experiments ("Http," n.d.).

TODO: Add the precise repository URL, release or commit identifier, command-line invocation, configuration file, dependency versions, trained model availability, and data-access statement. If the analysis code and model weights will be released only upon publication, state that explicitly.

3 Results

3.1 Cohort characteristics

The exploratory cohort comprised 45 participants distributed across five clinical groups: young healthy volunteers, older healthy volunteers, cystic fibrosis (CF), chronic obstructive pulmonary disease (COPD), and interstitial lung disease (ILD). All participants contributed a three-dimensional hyperpolarized xenon-129 ventilation image. Diagnostic labels were withheld from model training and were introduced only after latent encoding for evaluation of the learned representation.

TODO: Add a cohort table reporting the number of participants, age, sex, pulmonary-function measurements, ventilation defect percentage (VDP), and relevant clinical characteristics for each group. Report missing data explicitly and provide the corresponding omnibus and pairwise demographic comparisons.

3.2 Model optimization and reconstruction

The three-dimensional multiscale normalizing flow was trained by exact-likelihood optimization after spatial resampling and probabilistic dequantization of the ventilation volumes. Training remained numerically stable, and all 45 images were successfully mapped to the four-level latent representation and reconstructed through the analytic inverse. No diagnostic information was used to construct the latent space.

TODO: Report the selected checkpoint, training and validation bits per dimension, convergence behavior, and the criterion used for checkpoint selection. Add quantitative reconstruction error, expected to be near numerical precision for the dequantized model input, and distinguish this invertibility check from reconstruction of the original pre-dequantization image. Include a training-curve figure if it contributes information beyond the reported values.

Visual inspection indicated that reconstructed volumes retained the pulmonary signal distribution and subject-specific ventilation abnormalities present in their corresponding inputs. Because exact inversion alone does not establish that samples or interpolations occupy well-supported regions of the learned distribution, reconstruction fidelity was considered a verification of model implementation rather than evidence of generative validity.

TODO: Add representative input/reconstruction pairs from each clinical group using identical display windows and anatomical planes. If reconstruction differences are imperceptible, report a difference image with an appropriately amplified and explicitly labeled intensity scale.

3.3 Organization of the multiscale latent representation

Encoding produced four latent components, $\mathbf{z}_0$ through $\mathbf{z}_3$, corresponding to the successive levels of the multiscale architecture. The complete representation retained a bijective correspondence with the input image, whereas the level-specific components provided complementary descriptions of variation within the cohort. After radial projection, encoded subjects occupied the hyperspherical Gaussian typical set and could be compared without treating the low-probability latent origin as a representative population image.

Pairwise distances varied across resolution levels, demonstrating that subject relationships were not invariant to the level at which the representation was examined. Fine and coarse latent components produced different relative organizations of the cohort. The observed group relationships were consistent with the multiscale representation capturing complementary aspects of ventilation variation; however, architectural scale alone was insufficient to assign a specific biological interpretation to an individual level.

TODO: Report, for each level, the latent dimensionality, empirical radius before projection, radial dispersion, and distribution of pairwise distances. Add a matrix or heat map showing the subject-by-subject distances ordered by clinical group. Quantify the association between the level-specific distance matrices using Mantel or rank correlations with subject-label permutations. These results will establish whether the levels contain complementary information rather than relying on visual interpretation.

3.4 Image-domain interpretation of latent scale

Spherical interpolation provided continuous trajectories between subject representations while remaining on a common-radius hypersphere. Decoding intermediate points generated a corresponding sequence of three-dimensional ventilation images. These trajectories demonstrated the operational advantage of the invertible formulation: latent differences could be returned directly to the image domain rather than interpreted solely through a two-dimensional embedding.

TODO: Select prespecified subject pairs representing (1) two participants within the same group,

(2) a healthy-to-disease comparison, and (3) two disease groups with similar VDP. Show equally spaced Slerp points using fixed display parameters. To support claims that L_0/L_1 represent localized abnormalities and L_3 represents global organization, perform level-restricted latent replacement or interpolation while holding the other levels fixed. Report quantitative image changes at each level, such as spatial-frequency content, connected-component characteristics of low-ventilation regions, or regional displacement of ventilation signal.

3.5 Clinical organization of latent distances

Although clinical labels were absent during optimization, the distance matrices displayed group-associated structure after training. The strongest exploratory relationships involved comparisons between young healthy volunteers and disease groups and between the two obstructive disease groups, CF and COPD. Comparisons involving CF versus ILD and CF versus older healthy volunteers showed greater overlap. These observations suggest that the learned representation contains clinically relevant information beyond an undifferentiated measure of overall ventilation loss, while also revealing boundaries that remain ambiguous in the small exploratory cohort.

The previously calculated pairwise Welch tests yielded very small nominal p values for several comparisons. These values were not retained because distances sharing a subject are statistically dependent and cannot be treated as independent observations. Inferential results will instead be based on permutation of labels at the subject level.

TODO: Replace this paragraph with the final statistical results. For each of L_0–L_3 and the total distance, report:

- omnibus PERMANOVA pseudo-F, variance explained (R^2), and permutation-derived p value;
- PERMDISP statistic and permutation-derived p value;
- pairwise PERMANOVA effect sizes and multiplicity-adjusted p values;
- within-group and between-group distance summaries with confidence intervals; and
- sensitivity analyses demonstrating that significant location effects are not explained solely by unequal group dispersion.

With five groups, ten pairwise group comparisons were possible at each of the five distance definitions, yielding 50 planned tests and a Bonferroni family-wise threshold of 1.0×10^{-3}. Exact permutation counts and attainable p-value resolution should accompany the final results. A compact table should report all effect sizes and corrected p values, whereas the main text should emphasize only the comparisons that address the principal methodological hypotheses.

3.6 Comparison with ventilation defect percentage

The normalizing-flow representation and VDP encode fundamentally different quantities. VDP assigns each subject a single defect-burden value, whereas the flow produces an invertible multiscale representation from which spatially informed subject distances can be derived. The key empirical question is therefore not whether the latent distance reproduces VDP, but whether it distinguishes images that have comparable VDP yet different spatial organizations of ventilation abnormality.

TODO: Add the subject-level VDP values and perform the following prespecified analyses:

1. quantify associations between VDP differences and latent distances at each level using subject-label or matrix-based permutation testing;
2. identify subject pairs with similar VDP but large latent distance and show their ventilation images;
3. identify pairs with different VDP but relatively small latent distance to characterize potential failure modes; and
4. compare the ability of VDP and latent-distance features to recover clinical group structure using cross-validated or permutation-based metrics appropriate for the small sample.

These analyses are required to support the central claim that the flow retains clinically relevant spatial information discarded by the scalar summary. Without them, the comparison between VDP and the latent representation should remain conceptual rather than be described as demonstrated superiority.

3.7 Summary of findings

The experiments established the technical feasibility of exact-likelihood, invertible modeling of three-dimensional hyperpolarized ^{129}Xe ventilation MRI in a sparse functional-imaging setting. The learned multiscale representation supported reconstruction, level-specific subject comparison, and image-domain interpolation without diagnostic supervision. Exploratory organization of the latent distances was associated with clinical phenotype, particularly for young healthy versus diseased ventilation and for CF versus COPD, while other group boundaries remained less distinct. Final claims concerning statistical group separation, scale-specific biological interpretation, and added value relative to VDP await the subject-level permutation and image-domain analyses specified above.

4 Discussion and Conclusions

4.1 Principal findings

Collectively, this framework extends quantitative functional lung imaging beyond predefined voxel classes and one-dimensional defect summaries. It provides an invertible connection between population geometry and complete three-dimensional images, thereby supporting multiscale comparison and generative interrogation within a single probabilistic model. The current study is intended as a methodological proof of concept as it characterizes the representation, identifies the assumptions required for its interpretation, and establishes an experimental basis for subsequent validation in larger and more heterogeneous cohorts.

This study investigated three-dimensional normalizing flows as a quantitative framework for hyperpolarized xenon-129 (^{129}Xe) pulmonary ventilation MRI. The central methodological premise was that conventional scalar measurements and an invertible generative representation answer different questions. Ventilation defect percentage (VDP) estimates the total burden of signal classified as defective, whereas the normalizing flow retains a bijective correspondence between the complete ventilation image and a multiscale latent coordinate. The latter therefore provides access to spatial relationships between subjects, resolution-dependent variation, and image-domain trajectories that cannot be recovered from VDP alone.

The experiments established the technical feasibility of applying exact-likelihood, multiscale flow modeling to sparse three-dimensional functional lung images. Despite extensive background and highly quantized intensities, probabilistic dequantization enabled continuous-density estimation, and each ventilation volume could be represented across four latent resolution levels and reconstructed through the analytic inverse. The resulting representation supported pairwise comparison on the Gaussian typical set and spherical interpolation between subjects. Importantly, diagnostic labels were excluded from training. Clinical organization observed after encoding therefore reflects information present in the images and captured by the likelihood-based model rather than direct optimization of a classification objective.

Exploratory analyses suggested that the learned geometry contains group-associated information. The most apparent relationships involved young healthy volunteers versus disease groups and CF versus COPD, whereas CF versus ILD and CF versus older healthy volunteers showed greater overlap. These observations are consistent with the possibility that diseases producing similar overall ventilation impairment may differ in their spatial expression. Nevertheless, the current results should be interpreted as evidence of representational feasibility rather than definitive disease discrimination. Valid inference requires subject-level permutation analysis, and estimates from this small cohort will retain substantial uncertainty even after the dependence among pairwise distances is handled correctly.

4.2 Relationship to conventional ventilation quantification

VDP remains an important and clinically useful measure. It is readily interpretable, can be calculated using several established segmentation procedures, and has demonstrated reproducibility and potential utility in multicenter and interventional settings (Couch et al., 2019; Svenningsen et al., 2020). The present framework is therefore not motivated by the claim that VDP is intrinsically erroneous. Rather, VDP is intentionally reductive: it maps a three-dimensional image to a single proportion. That reduction is advantageous when a simple measure of defect burden is desired, but it is insufficient when the number, location, topology, or regional organization of abnormalities is relevant.

This distinction extends our previous comparison of histogram- and image-based quantification (Tustison et al., 2021). That work showed that optimization in the image domain uses spatial context discarded by histogram-based approaches and can improve measurement precision in the presence of common MRI perturbations. However, even an image-based segmentation is typically summarized by VDP at the final stage. The current work moves the point of comparison from segmentation algorithm to representation: instead of asking how best to assign every voxel to a predefined class, it asks whether the complete image can be embedded in a probabilistic coordinate system without an irreversible intermediate labeling.

The two approaches should consequently be regarded as complementary. VDP provides a concise measure with direct clinical meaning, whereas latent distances provide a higher-dimensional description whose interpretation must be established empirically. A critical next analysis within the current cohort is to identify subjects with comparable VDP but divergent latent representations and determine whether their images exhibit meaningful spatial differences. Conversely, cases with different VDP but small latent distance will reveal whether the learned geometry discounts changes in total burden that remain clinically important. Such discordant pairs will be more informative than a simple correlation between VDP and latent distance because they directly characterize the information preserved or deemphasized by each representation.

4.3 Interpretation of the multiscale latent space

The multiscale construction is a principal advantage of the Glow-derived architecture (Kingma and Dhariwal, 2018). Factoring latent variables across successive spatial resolutions creates several related coordinate systems while preserving the invertibility of their combination. The preliminary group relationships differed across L_0–L_3, suggesting that the levels are not redundant. However, it would be premature to equate individual levels directly with specific biological scales. A finer architectural resolution does not necessarily encode only small ventilation defects, nor must a deeper level exclusively describe global pulmonary organization. Coupling transformations preceding each split permit information to be redistributed across channels and levels.

Biological interpretation therefore requires intervention in the latent representation followed by decoding. Level-restricted interpolation, replacement, or perturbation while holding the remaining components fixed can reveal the image features controlled by each component. Spatial-frequency analysis, regional summaries, and connected-component characteristics of low-ventilation regions can then quantify whether the decoded effects are local, lobar, or global. This image-domain validation is essential if the multiscale factorization is to support more than an architectural description.

Similar caution applies to the latent distance. A normalizing flow supplies a differentiable bijection and tractable probability model (Dinh et al., 2017; Kobyzev et al., 2021; Papamakarios et al., 2021), but it does not guarantee that Euclidean or hyperspherical proximity corresponds to biological similarity. The Gaussian base distribution constrains the aggregate latent density, whereas the learned transformation determines how image variation is arranged within that density. Spherical interpolation avoids trajectories through the low-probability origin and respects the radial concentration of high-dimensional Gaussian samples, but the resulting arc length remains model dependent. Its clinical meaning must be supported through external variables, decoded trajectories, stability across training seeds, and validation in independent data (Arvanitidis et al., 2018; White, 2016).

4.4 Statistical considerations

Pairwise distances create a nonstandard inferential setting because the 45×45 distance matrix contains far more entries than independent subjects. Distances that share a subject are correlated;

treating them as independent observations inflates the apparent sample size and can yield severely anti-conservative p values. For this reason, the preliminary Welch tests applied directly to distance entries should not be used to support group separation.

The appropriate exchangeable unit is the subject. PERMANOVA with subject-label permutations provides an omnibus test of group-associated location differences in the distance geometry, while PERMDISP evaluates whether apparent separation may instead reflect unequal within-group dispersion. Both are needed because disease cohorts may be intrinsically more heterogeneous than healthy cohorts. Pairwise permutation tests can subsequently localize effects, with multiplicity correction across clinical comparisons and resolution levels. Effect sizes and uncertainty intervals should be emphasized over significance alone, particularly in this exploratory cohort.

The statistical question should also remain distinct from classification. A significant difference in group geometry does not imply perfect separation, diagnostic accuracy, or clinical utility. Demonstrating those properties would require a prespecified prediction task, subject-level cross-validation, comparison with appropriate baselines, and external validation. The present study instead evaluates whether an unsupervised image representation contains group-associated structure, which is a necessary but not sufficient condition for subsequent biomarker development.

4.5 Limitations and future work

Several limitations remain. First, the cohort of 45 subjects is small relative to the dimensionality and capacity of the three-dimensional flow. The model may encode cohort-specific acquisition characteristics or sampling variation in addition to pulmonary biology. Group sizes and demographic distributions may also be unbalanced, and age is partly confounded with clinical grouping when young and older healthy volunteers are treated separately. Larger independent cohorts will be required to estimate disease heterogeneity, adjust for demographic and technical covariates, and assess generalization across acquisition protocols.

Second, the current model is monocontrast and uses only ventilation MRI. This design isolates the functional signal and avoids dependence on a paired structural acquisition, but it limits anatomical contextualization. Pulmonary boundaries, body habitus, lung volume, positioning, and registration may contribute to latent distance. Image-domain experiments and sensitivity analyses using masks, alternative spatial normalization procedures, and nuisance regression will be required to determine how much of the geometry is attributable to ventilation distribution rather than global shape or preprocessing.

Third, dequantization is necessary for continuous likelihood modeling but introduces a tunable stochastic perturbation. The selected scale of 0.15 should be justified through sensitivity analysis. Too little dequantization may leave discrete intensity structure that is poorly matched to the density model, whereas excessive noise may obscure small or low-contrast ventilation abnormalities. Comparisons across dequantization levels should therefore include likelihood, sampling behavior, latent-distance stability, and image-domain preservation of clinically relevant structure (Ho et al., 2019).

Fourth, exact-likelihood image flows impose substantial memory and computational costs because invertibility and Jacobian evaluation require retaining or recomputing intermediate transformations. These demands currently constrain spatial resolution and network capacity for three-dimensional experiments. Resampling to $48 \times 32 \times 48$ voxels makes the present analysis tractable but may attenuate small peripheral defects. Future work should examine memory-efficient invertible architectures, patch- or region-based strategies, and higher-resolution models while ensuring that changes in architecture do not compromise the comparability of latent distances.

Fifth, likelihood and perceptual or clinical fidelity are not equivalent. A model can assign favorable likelihood while producing samples or trajectories that fail to preserve subtle disease features. Evaluation must therefore combine likelihood with reconstruction checks, decoded interpolation experiments, spatial measurements, expert review, and task-specific validation. Stability across checkpoints and random seeds is particularly important because the orientation and local arrangement of a Gaussian latent space are not uniquely identifiable.

Finally, the present work does not establish superiority to VDP, clinical diagnostic performance, or generalization to a multicenter population. Those claims require direct empirical comparisons and substantially larger data. The future analysis of large multicenter collections should be treated as a separate validation stage rather than as evidence supporting the current cohort. Such studies can determine whether the framework captures reproducible disease-related structure after accounting for site, scanner, acquisition protocol, and demographic variation.

4.6 Software availability and reproducibility

The methodological contribution is accompanied by a mature, publicly available implementation within the ANTsX/ANTsTorch ecosystem ("Http," n.d.). Open implementation is especially important for a framework whose behavior depends on preprocessing, dequantization, multiscale architecture, checkpoint selection, and distance construction. Release of the training configuration, model weights, exact latent-distance code, and subject-level statistical workflow will permit others to reproduce the reported geometry and test alternative modeling assumptions. Reproducibility should include not only the final checkpoint but also the parameters required to regenerate preprocessing and evaluation outputs.

4.7 Conclusions

Multiscale three-dimensional normalizing flows provide an invertible probabilistic framework for quantitative analysis of hyperpolarized ^{129}Xe pulmonary ventilation MRI. By retaining a direct correspondence between complete images and a structured Gaussian latent representation, the approach extends functional lung quantification beyond predefined voxel classes and scalar defect burden. The framework supports reconstruction, resolution-specific analysis, subject-to-subject distance measurement, and image-domain interrogation of latent trajectories without using diagnostic supervision during training.

In the present exploratory cohort, the learned geometry exhibited clinically associated organization, but its statistical strength, scale-specific biological meaning, and added value relative to VDP require the planned subject-level permutation and image-domain analyses. Accordingly, the principal contribution of this study is not a finalized diagnostic biomarker but a technically grounded and reproducible framework for studying spatial variation in pulmonary ventilation. With rigorous validation in larger independent cohorts, this representation may complement conventional defect summaries and support more detailed phenotyping of functional lung disease.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used Gemini (Google) in order to refine the manuscript's structure, improve grammatical clarity, assist in the translation of technical sections, and conduct preliminary literature scans via deep research tools. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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